



UnionPay Integrated Circuit Card Specifications

— Product Specifications

Part VIII Display Card Specification



Version 2014

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1 Application Scope

This book applies to all UPI participants.

2 References

The following normative documents contain provisions which, through reference in this text constitute provisions of this Part. For dated references, subsequent amendments to, or revisions of, any of these publications (excluding corrected contents) do not apply. However, parties to the agreements based on this Part are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies.

ISO/IEC 7816-1:1987—*Identification cards -- Integrated circuit(s) cards with contacts -- Part 1: Physical characteristics*

ISO/IEC 7816-2:1988—*Identification cards -- Integrated circuit(s) cards with contacts -- Part 2: Dimensions and location of the contacts*

ISO/IEC 7816-3:1989—*Identification cards -- Integrated circuit(s) cards with contacts -- Part 3: Electronic signals and transmission protocols*

ISO/IEC 7816-4—2005 *Identification cards -- Integrated circuit cards -- Part 4: Organization, security and commands for interchange*

ISO/IEC 7816-5—1994/Amd 1:1996 *Identification cards -- Integrated circuit(s) cards with contacts -- Part 5: Numbering system and registration procedure for application identifiers*

ISO/IEC 7816-6—1996 *Identification cards -- Integrated circuit(s) cards with contacts -- Part 6: Interindustry data elements*

ISO 4217—2001 *Codes for the representation of currencies and funds*

3 Terms and Definitions

The following terms and definitions are applicable to this Part.

3.1 Boost Circuit

The boost circuit is also called bootstrap circuit, which exploits the bootstrap diode, bootstrap capacitance and other electronic elements to superimpose the discharge voltage of the capacitance and the source voltage so as to boost the voltage. Sometimes, the boosted voltage may be several times the source voltage.

3.2 E Paper

The technology that offers comfortable paper-like reading experience with thin, light and bendable media and low power consumption is generally called E-paper technology; the E paper refers to the real-paper-like electronic display.

3.3 LCD Liquid Crystal Display

Is a type of thin flat panel display that uses electro-optical features of the liquid crystal to produce letters, images and videos, etc.

3.4 Terms on Requirement

This Specification has different requirements for conformance, with relevant terms defined as follows:

“Shall” indicates requirements to be followed are mandatory;

“Should” indicates a recommended possibility;

“May” means choices may be freely made.

The following symbols shall be adopted when tags appear in commands or responses:

M: mandatory tag shall appear

C: conditional tag shall appear under certain conditions

O: optional tag may or may not appear.

3.5 Cards

“Cards” in this Specification refer to payment cards that have the same size as the traditional credit cards and the contactless payment applications and may be interactive with the payment reader.

Expression of Numbers

The following symbols are adopted in this Specification:

The decimal number is expressed by the number itself (for example: Number 1)

The hexadecimal numbers is expressed by the number within a quotation mark, and with an x in front of it (for example: number x ‘01’).

The binary number is expressed by the number within a quotation mark, and with a b in front of it (for example: b'00000001' or b'1').

Bit 1 means the least significant bit (0x01); Bit 8 means the most significant bit (0x80)

3.6 RFU (reserved) Data Coding

The coding of data with RFU symbols shall be carried out in compliance with the same principle in the UnionPay Integrated Circuit Card Specification.

4 Symbols and Abbreviations

The following abbreviations and symbols are applicable to this Part.

AAC	Application Authentication Cryptogram
AC	Application Cryptogram
AID	Application Identifier
ARQC	Authorization Request Cryptogram
DDA	Dynamic Data Authentication—offline data authentication may prevent the use of false cards. The RSA signature generated by the card based on particular data during a transaction is used for terminal verification.
DES	Data Encryption Standard
GPO	Get Processing Options
Hex	Hexadecimal
IC	Integrated Circuit
ICC	Integrated Circuit Card
LCD	Liquid Crystal Display
IDD	Issuer Defined Data
ISO	International Standardization Organization
MAC	Message Authentication Code
UICS	UICS Debit/Credit Application. UICS in this Specification refers to applications that support the contact interface in full compliance with the requirements of EMV 4.1.
PIN	Personal Identification Number
PPSE	Proximity Payment Systems Environment---a list of supported application identifiers, application tags and the application priority indicator, which may be accessed via the contactless interface. The list, including all directory entries, is returned in FIC in response to the SELECT PPSE ('2PAY.SYS.DDF01') command.

PSE	Payment System Environment
q UICS	Contactless Quick ICS---to ensure quick transactions via the contactless interface. The requirements for such transactions are described in this Specification.
RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RSA	A type of asymmetric cryptographic algorithm developed by Rives, Shamir and Adleman for encryption and authentication.
TC	Transaction Certificate-- the application cryptogram generated during transaction acceptance.
TLV	Tag, Length and Value
TN LCD	Twisted Nematic Liquid Crystal Display
STN LCD	Super Twisted Nematic Liquid Crystal Display (in short, STN)
RoHS	A compulsory standard formulated by European Union; its full name is Restriction of Hazardous Substances in Electronic and Electrical Equipment.

5 Overview

5.1 Definition of the Financial IC Display Cards

The financial IC display cards (hereinafter referred to as Display Cards) are a kind of value-added products of the financial IC cards, which may display the non-sensitive information within the display cards, such as EC balance, transaction log and custom data of the issuer by integrating the E paper technology and other display technologies and may also display the transaction authentication data by combining dynamic passwords, the digital certificate application and other functions.

The display cards have to be dual interface cards. The financial IC chip shall support the debit and credit applications and the micropayment function of the standard CUP IC cards, and shall also be able to support and display communications with the display module. If the display module does not work, while the chip of the financial IC card functions well, the chip shall be able to work continually.

5.2 Working Principle of Financial IC Display Cards

The functional framework of display cards is shown as follows:

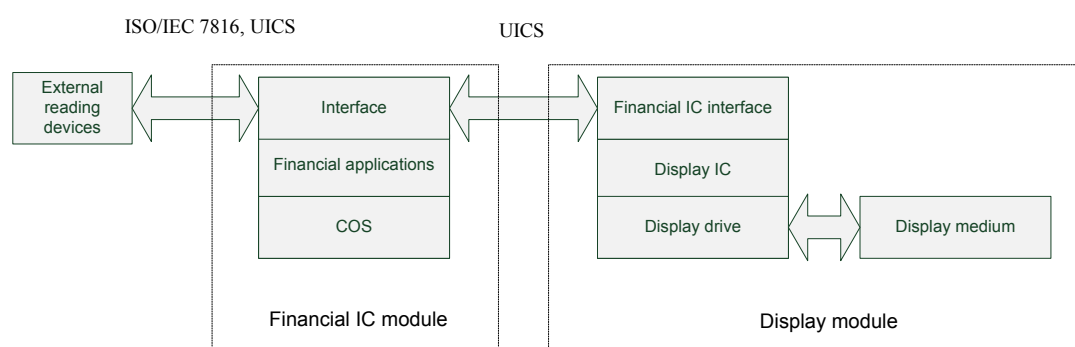


Fig. 5-1 Functional Framework of Display Card

A display card is composed of two major functional modules: the financial IC module and the display module. The financial IC module shall function in accordance with the requirements of UICS, and shall at the same time, have a communication interface with the display module; the display module shall be able to obtain specified data from the financial IC module and display such data via the display medium according to conventions of internal procedures and external inputs.

The working mode of the financial IC display cards may be divided into two types according to the difference of interface between the financial IC module and the display module.

Active mode of the financial IC module: the financial IC chip actively pushes specified data during a transaction to the display module after being connected with power via POS or ATM or other external devices. The display module will directly display such data on the display medium or firstly store them in the storage area of

the display module and display such data after receiving corresponding keyboard inputs of the display card. The active mode of the financial IC module is applicable to the push of non-sensitive, static data, such as EC balance and data specified by the issuer. In this mode, the financial IC chip is required to actively transfer the specified data according to a specific format to the display module via the interface with the display module; refer to Part 9 for definition of the format.

Active mode of the display module: the display module actively sends commands to the financial IC module and read and display the specified data from data returned from the financial IC module. In this mode, the power of the financial IC display card is supplied by a battery within the card and the display function is triggered by the keyboard input of the card. This mode may be used to read the non-sensitive, static data such as EC balance and specified data of the issuer, and may also be used to generate CUP standard dynamic passwords. Relevant procedures and commands shall be consistent with UICS.

6 Electro-Mechanical Properties, Logical Interface and Transport Protocol

The internal physical structure of the display cards defined in this Specification is composed by the financial IC module and the display module.

The financial IC module includes a financial IC chip, a contact interface circuit and a contactless interface circuit. The display module includes a display chip, a display driving circuit, a keyboard I/O and a display medium. The display module using E paper as the display medium also includes a boost circuit. The display medium may be E paper or LCD, etc. A battery is used to supply power to the display module and the financial IC chip in order to obtain and display data in the financial IC chip. The users may use the keyboard to select functions of the financial IC display card and input transaction information. The communication between the financial IC module and the display module is conducted via the ISO/IEC 7816 or other interfaces.

Fig. 6-1 shows the hardware composition of the financial IC display cards. In the Figure, the keyboard and power are optional elements.

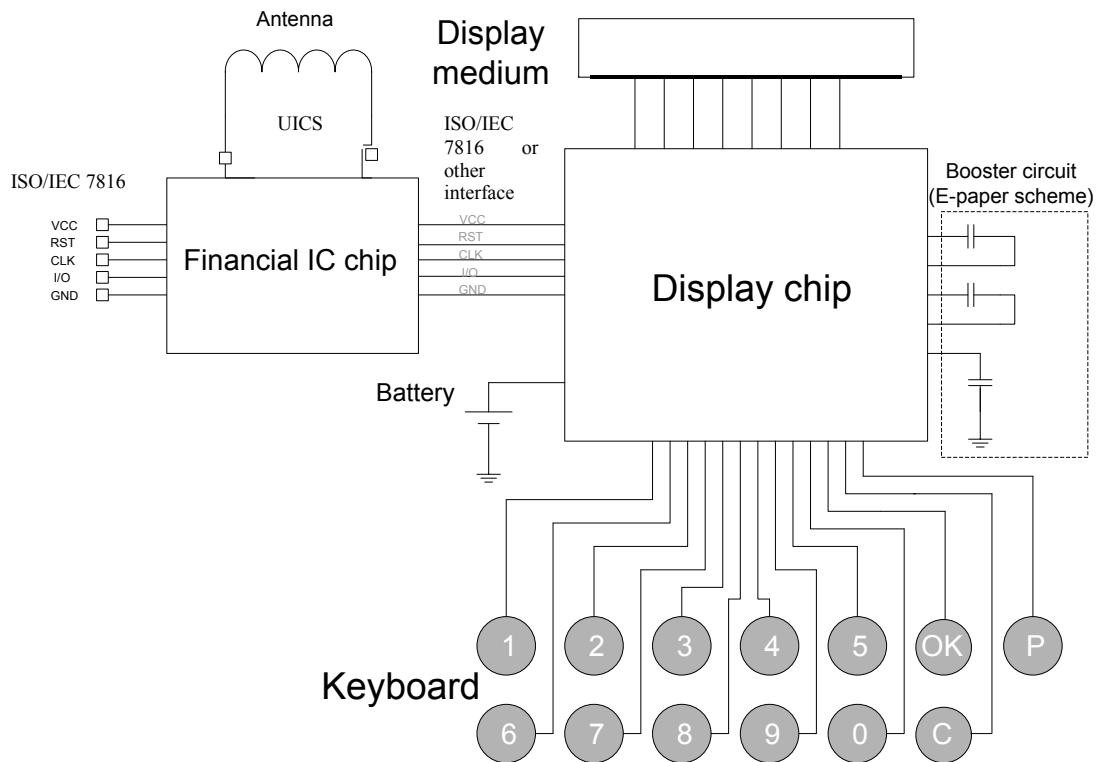


Fig. 6-1 Physical Structure of Financial IC Display Cards

6.1 Electro-Mechanical Properties, Logical Interface and Transport Protocol of Financial IC

The electro-mechanical properties, logical interface and transport protocol of the financial IC shall meet relevant requirements for the display cards.

6.1.1 Power Consumption Requirements for Financial IC

If the display card is battery-powered, the financial IC card chip shall adopt low power consumption and leak-proof designs, and the chip power consumption and battery supply shall meet following requirements:

If the financial IC chip is under the work mode and is battery-powered, the working current of the chip shall be controlled within 3mA; if the financial IC chip is not under work mode, the internal current of the display card shall be lower than 0.1uA.

6.2 Specification Requirements for the Display Medium

The display medium of single-currency EC financial IC display cards shall at least support the display of 8-digit letters or numbers, and shall be able to display the decimal point “.” before the second letter or number from right to left.

The display medium of multi-currency EC financial IC display cards shall at least support the display of 11-digit letters or numbers, and shall be able to display the decimal point “.” before the second, third and fourth letter or number from right to left.

The display medium includes E paper and LCD. The requirements for their functions and physical properties are presented as follows.

6.2.1 Specification Requirements for E Paper

6.2.1.1 Display Content Requirements

The E paper adopting segment codes shall at least support 7-segment codes of each character and be able to display 10-digit numbers in the range of 0-9 and the decimal point. The E paper adopting 14 segment codes shall support 10-digit numbers in the range of 0-9, capital letters in English and the decimal point.

6.2.1.2 Electrical Property Requirements

Working voltage: 2.0V~5.5V; Quiescent Current: lower than 100nA

Viewing angle of the display: full view

White reflectivity: greater than 30% under indoor temperature of 25°C

Contrast ratio: above 8:1 under indoor temperature of 25°C, with the minimum no less than 6:1.

Response time: 240ms under indoor temperature of 25°C, with maximum no more than 400ms.

Display hold: more than 30 days.

Display compression resistance: 3 kgf/cm²

Display working temperature: -10°C~50°C

Bending features: 2000 times in each of four directions (conditions: indoor temperature 25°C, conforming to the ISO 10373 : IC Card Bending Test Conditions).

6.2.2 Specification Requirements for LCD

6.2.2.1 Display Content Requirements

The LCD has two types: segment code type and dot matrix type.

The 7-segment LCD shall be able to display 10-digit numbers in the range 0-9 and decimal point; the 14-segment LCD shall be able to display 10-digit numbers in the range of 0-9, capital letters in English and decimal point.

The dot matrix LCD shall at least support 16×64 points, and be able to display numbers, English characters, decimal point and Chinese characters.

6.2.2.2 Electrical Property Requirements

Display Type: TN

Quiescent current under non-display status: 0.1uA

Drive current under display status: <3.0uA

Viewing direction of the display: 12H

Viewing angle of the display: ±45°

Contrast ratio: 10:1 (conditions: indoor temperature 25°C, driven under 3.0V)

Response time: 60ms, with the maximum no more than 100ms (conditions: indoor temperature 25°C, driven under 3.0V)

Display compression resistance: 20 kgf/cm²(conditions: indoor temperature 25°C, plane compression, instead of pot or linear compression).

Display working temperature: -20°C~50°C

Display storage temperature: -30°C~60°C

Display working humidity: 20%~85%

Display storage humidity: 5%~95%

Anti-static: 4KV (contact)/9KV (air-isolated)

Bending features: 2000 times in each of four directions (conditions: indoor temperature 25°C, conforming to the ISO 10373 IC: Card Bending Test Conditions).

6.3 Specification Requirements for Battery

The display cards meeting the requirements of this Specification shall be able to work for more than 3 years or supporting at least 7000 times of active operations. The active operations are defined to be financial IC chip operations completed based on power supplied by the battery. The battery used in display cards shall meet the above requirements.

For lithium batteries used in the display cards, they shall meet the following requirements:

Rated voltage: 3.0 V

Nominal battery capacity: above 14mAh (indoor temperature 23°C)

Maximum-continuous discharge (23°C): $C / 2$ (C refers to the battery capacity; the maximum discharge current in this list is $C/2A$) If the battery capacity is 14mAh, the maximum discharge current is $C/2=14mA/2=7mA$.)

Working temperature: -10°C~60°C

Storage temperature: -20°C~40°C

Maximum thickness: 0.45mm

Self-discharge rate: <1 %per month

Safety: RoHS

Service life: more than 3 years.

Note: the usage times of cards using a lithium battery is calculated based on the following: usage times of cards= (battery capacity-(self-discharge of the battery × 3 years + system standby power consumption × 3 years))/power consumption each time of usage

6.4 Specification Requirements for Keys

The display card may be with a single key or multiple keys or without key. Specification requirements for keys:

The single-key display cards include one key, which may turn on or turn off the power and act as a function key.

The multi-key display cards include number keys from 1 to 9, a confirm key, a delete/mode key and other function keys as well as a power key. The power key shall be a mechanical key to avoid mistaken operations.

6.4.1 Pressure Requirements for Keys

The minimum working pressures of keys is 1N. When the external pressure is smaller than 1N, the keys will not respond in order to avoid mistaken operations. It shall be ensured that the keys will not be harmed under maximum working pressure of 50N.

6.4.2 Fatigue Strength of Keys

All keys of the financial IC display cards are required to pass 50,000 times of fatigue tests. The touch keys shall support touch operations of more than 50,000 times.

6.4.3 Height of Key Modules

The highest point of key modules shall not be 0.10mm higher than the card surface; the thickness of cards containing key modules shall fall within the range required in UICS.

6.5 Interface and Transport Protocol between the Financial IC Chip and the Display Chip

Under the active mode of the display module, the interface and transport protocol between the financial IC chip and the display chip shall meet the requirements in Part 1-3 of ISO/IEC 7816 and requirements in UICS. The interface and transport protocol between the financial IC chip and the display chip under the active mode of the financial IC module are not specifically defined in this Specification. No matter which type of interface or transport protocol is used, the usage and security of the financial IC functions defined in UICS shall not be influenced.

7 Data Display Procedures under Active Mode of Financial IC Chip

The active mode of the financial IC chip is a type of working mode under which, the financial IC chip actively pushes the data in the card to be changed by the external contact or contactless POS, ATM or any other card reader to the display module.

Under the active mode, the batter-powered financial IC chip with function keys shall be able to store data in the display module, which may be accessed by the card holder via function keys.

The display cards without function keys shall be able to immediately display data after the financial IC chip has actively pushed such data to the display module. Refer to Part 9 of this Specification for relevant commands and definitions of data elements of the financial IC chip under active mode.

The display time length of data under active mode of the financial IC chip is a continuous, configurable time range and may be configured according to the needs of the issuer.

Time interval for time range: 0-128seconds

Default time length: 30seconds

The E-paper display cards that only support the function of displaying EC balance may choose to continuously display the current EC balance without disappearing.

7.1 Balance Display Procedures under Active Mode of Financial IC Chip

After the EC balance in the display card has been changed, the financial IC chip will push the changed balance to the display module, which will then display the data on the display medium. The financial IC chip of display cards under the active mode shall boast following functions:

With respect to battery-powered display cards with keys, after the E balance has been changed (following the completion of consumption, loading, unloading or other transactions) the financial IC chip shall be able to actively push the changed EC balance to the display module and display such data stored in the display module as well as read the data from the display module.

With respect to battery-powered multi-currency display cards with keys, after the E balance has been changed, the financial IC chip shall be able to actively push the currency types and the changed EC balance to the display module and display the currency types and EC balance data stored in the display module as well as read the currency types and the balance from the display module.

With respect to display cards without keys, after the EC balance in the display card has been changed, the financial IC chip shall be able to actively push the changed balance to the display module, which will then immediately display such data.

With respect to multi-currency display cards without keys, after the EC balance has been changed, the financial IC chip shall be able to actively push the currency types and the changed balance to the display module, which will then immediately display such data.

The multi-currency EC cards shall be able to display the currency types and the EC balance, refer to ISO 4217:2001 *Codes for the representation of currencies and funds*; for example, RMB is expressed as CNY, and U.S dollars, USD.

Following data, as shown in Table 7-1 shall be applicable to the currency types and EC balance defined in this Specification.

Name of Data Elements:	Tag	Length	Format
Application Currency Code	9F51	2	n4
EC Balance	9F79	6	n12
Application Currency Code of EC in the Second Currency	DF71	2	n4
EC Balance in the Second Currency	DF79	6	n12

Table 7-1

The display time length of the currency type and EC balance under active balance push mode of the financial IC chip is a continuous, configurable time range and may be configured according to the issuer's needs .

Time interval for time range: 0-128seconds

Default time length: 30seconds

The display cards that only support the function of displaying the EC balance may choose to continuously display the current EC balance without disappearing.

7.1.1 EC Balance Application Procedures under Active Push Mode of Financial IC Chip

The following procedures are applicable to transactions involving the change of EC balance, including EC consumptions, loading/unloading transactions, EC consumptions and loading/unloading transactions in multiple currencies, transactions with fees charged on multi-stage basis in extensive applications of the contactless IC card micropayment, offline pre-authorization transactions and offline authorization completion transactions that occurred at contact and contactless terminals.

For transactions without change of EC balance, the EC balance shall not be updated and displayed.

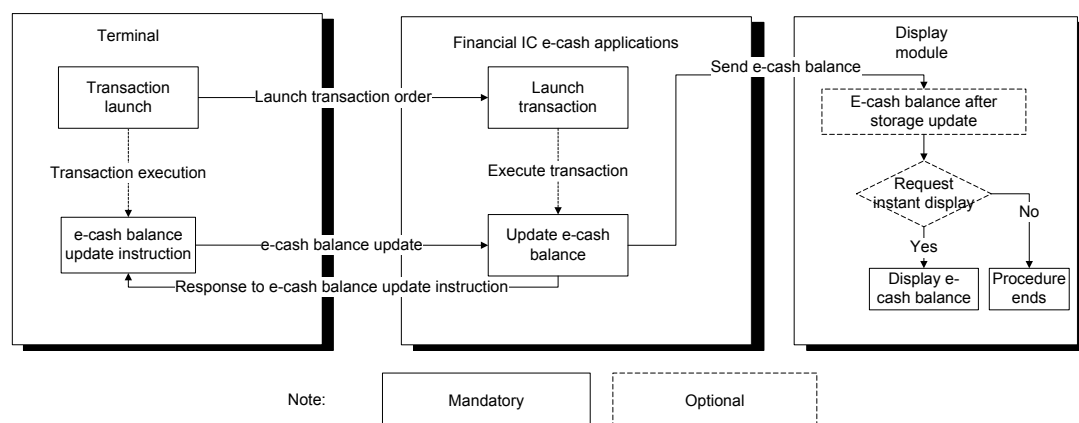


Fig. 7-1

1. The terminal sends the transaction start command to the card;
2. The terminal and the card execute the command;
3. The terminal sends the EC balance update command to the card, such as the Read Record command in offline EC consumptions and the scrip command of the issuer in loading transactions;
4. The financial IC executes the command by updating the EC balance;
5. The financial IC sends the updated EC balance to the display module;
6. After the display module has received the command, the battery-powered display card will update the EC balance stored in the display chip; the display card without keys or battery may not store the balance.
7. The display module will immediately display the updated balance or end the procedures according to the command.

7.1.2 Application Procedures for Displaying Balance via Keys

The battery-powered financial IC chip with function keys under the active mode may store the EC balance in the display module, which may be accessed by the card holder by choosing function keys and may be displayed on the display medium. The multi-currency EC display cards shall be able to display all currencies and corresponding EC balance via function keys.

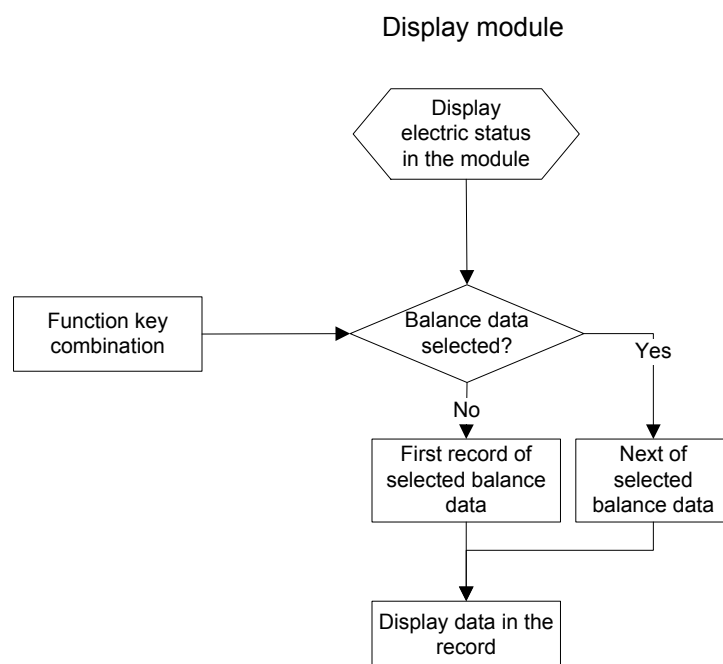


Fig. 7-2

Application Procedures for Displaying Balance via Keys:

1. If the display module is under the power-off state, after the user has pressed the power key, the display module will be changed to the power-on state;
2. In case the display module is under the power-on state and the balance display function has not been selected, if the user presses the balance display function key, the currency type and balance of the first currency will be displayed; in case the balance display function has been chosen, if the user presses the same function key, the currency type and balance of the second currency will be selected and displayed.

7.2 Transaction Log Display Procedures under Active Push Mode of Financial IC Chip

After the transaction log within the card has been updated, the financial IC chip will actively send the updated data to the display module. The display module will store the latest transaction data. The card holder may press the transaction log function key to select and display the latest record.

The transaction data shall be stored on a cyclic basis. The number of transaction records stored in the display module shall be no less than that in the financial IC. If the transaction log display function has been selected, the next record may be displayed via a key combination on a cyclic basis. The first record will be displayed following the last record. If the data are too long to be displayed in a row, the remaining data may be displayed by selecting the scrolling display via function keys of the display cards that use LCD as their display medium. For display cards that

use E paper as their display medium, the remaining data may be displayed on the next page via function keys.

The function of the financial IC chip to actively push the transaction log is defined to be an optional function in this Specification.

7.2.1 Application Procedures

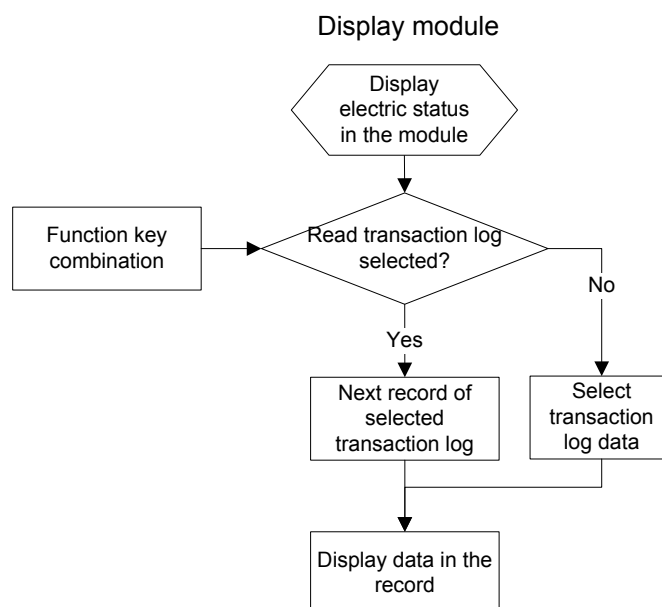


Fig. 7-3

1. The display module of battery-powered display cards updates the stored transaction log in manner of cyclic recoding.
2. After the display card has entered the power-on state, if the key combination for displaying the transaction log is selected, the latest record in the log will be selected and displayed.
3. If the transaction data display function has been selected, the user may read the next record on a cyclic basis via a key combination.

7.3 General Data Display Procedures under Active Push Mode of Financial IC Chip

In addition to the EC balance and the transaction log, the financial IC chip may actively push the specified data to the display module after such data have been updated according to needs of the issuer or after the financial IP chip has executed a command for data inquiry.

The function of the financial IC chip to actively push the general data is defined to be an optional function in this Specification.

7.3.1 Application Procedures

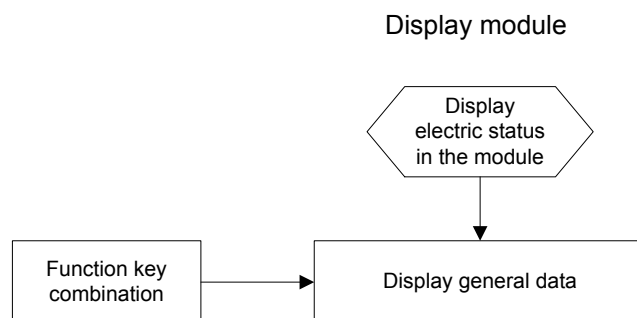


Fig. 7-4

1. The display module of the display cards with a battery updates the stored general data;
2. After the display card has entered the power-on state, the data may be displayed by selecting a corresponding key combination.

8 Data Display Procedures under Active Mode of Display Chip

This part defines the procedures for the display chip under active mode to read and display non-sensitive data in the financial IC chip via the display medium. The interface, protocol and APDUC commands between the display chip and the financial chip shall be consistent with definitions in UICS and are not detailed here.

When the display chip reads the financial IC chip, the card is powered by the internal battery. Data read by the display chip will be directly displayed on the display medium and will not be stored in the display chip.

The display time length of data under active mode of the display chip is a continuous, configurable time range and may be configured according to the needs of the issuer.

Time interval for time range: 0-128seconds

Default time length: 30seconds

The display cards that only support the function of displaying the EC balance may choose to continuously display the current EC balance without disappearing.

8.1 EC Balance Display Procedures under Active Mode of Display Chip

After the financial IC display card has been powered, if the function key for displaying the balance is selected, the display chip will send the APDU command to the financial IC to display the balance on the display medium.

The multi-currency EC display cards shall be able to display all currency types and balance via function keys.

The essential function of display cards that have keyboard display module is that their display chip is able to read the EC balance when under active mode.

8.1.1 Application Procedures

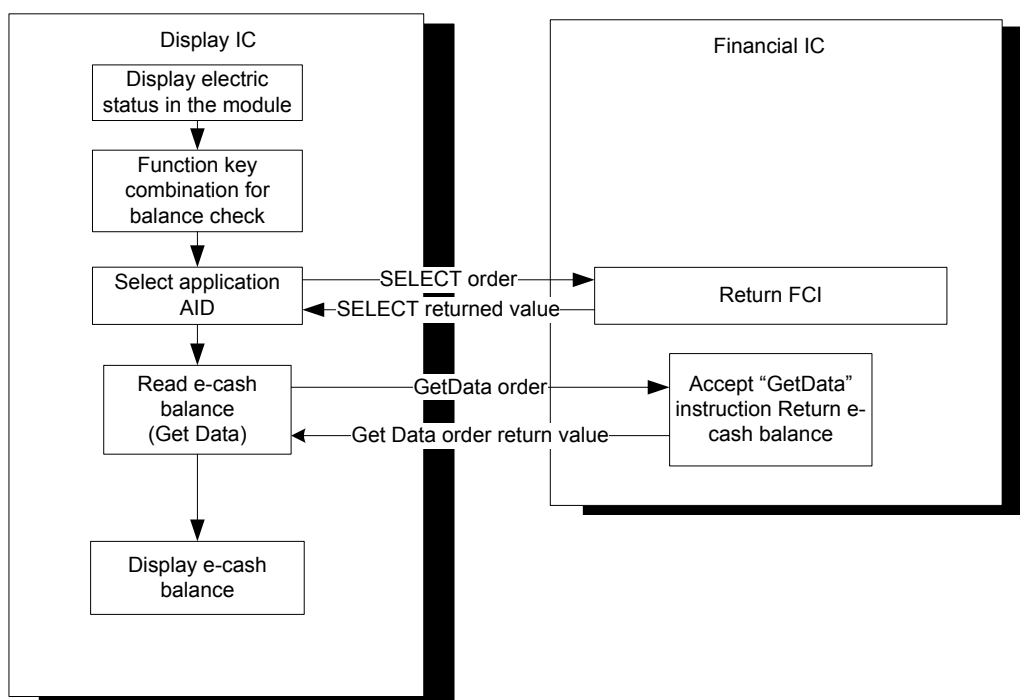


Fig. 8-1

1. If the display module is under the power-off state, after the user has pressed the power key, the display module will enter the power-on state;
2. If the display module has been under the power-on state, the user may press the key combination for balance display;
3. The display chip selects an application;
4. The display chips reads the EC balance by sending the Get Data command;
5. The display chip will send the EC balance to the display medium for display.

8.2 Transaction Log Display Procedures under Active Mode of Display Chip

After the display card has been powered, if the key combination for transaction log display is selected, the display chip will send the APDU command to the financial IC to display the transaction log. If the transaction log display function has not been selected, the latest transaction record will be displayed; if the function of displaying past transaction records has been selected, the next transaction record may be displayed on a cyclic basis via a key combination. The first record will be displayed following the last record.

The function of the display chip to read and display the transaction log is defined to be an optional function of the display cards in this Specification.

8.2.1 Application Procedures

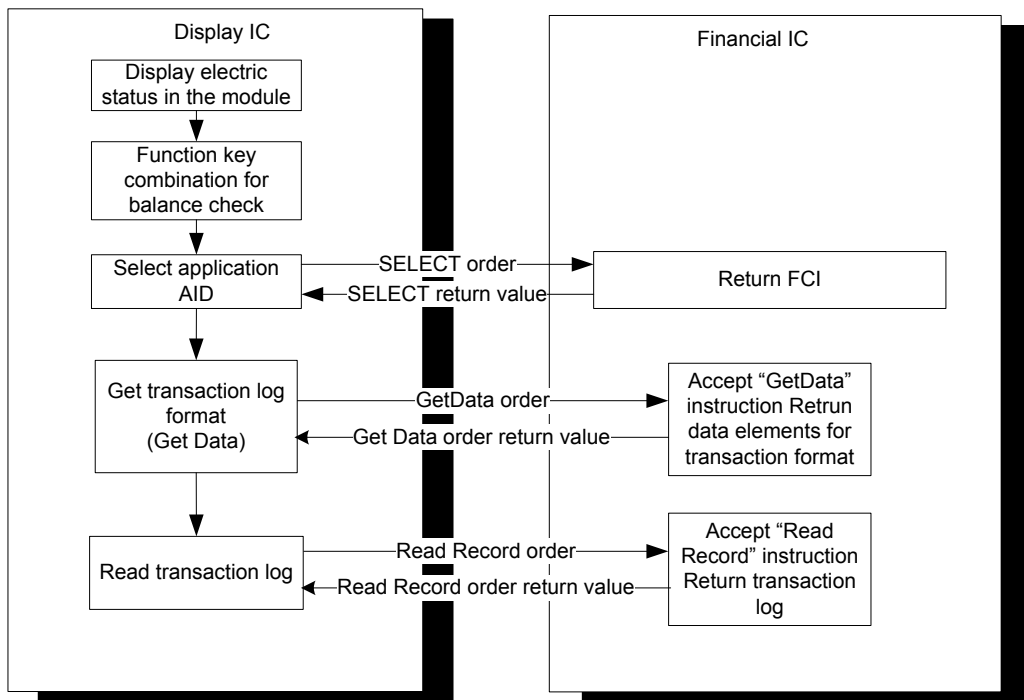


Fig. 8-2

1. If the display module is under the power-off state, after the user has pressed the power key, the display module will enter the power-on state;
2. If the display module has been under the power-on state, the user may press the key combination for transaction log display;
3. The display chip selects an application;
4. The display chips reads the data format of the transaction log by sending the Get Data command;
5. The display chips reads the transaction record by sending the Read Record command;

If the function of displaying past transaction data has been selected, the user may read the next record on a cyclic basis via a key combination.

8.3 General Data Display Procedures under Active Mode of Display Chip

In case the display card is under the power-on state, the display chip may read the data specified by the issuer and display them on the display medium.

The function of the display chip to read general data is defined to be an optional function of the display cards in this Specification.

8.3.1 Application Procedures

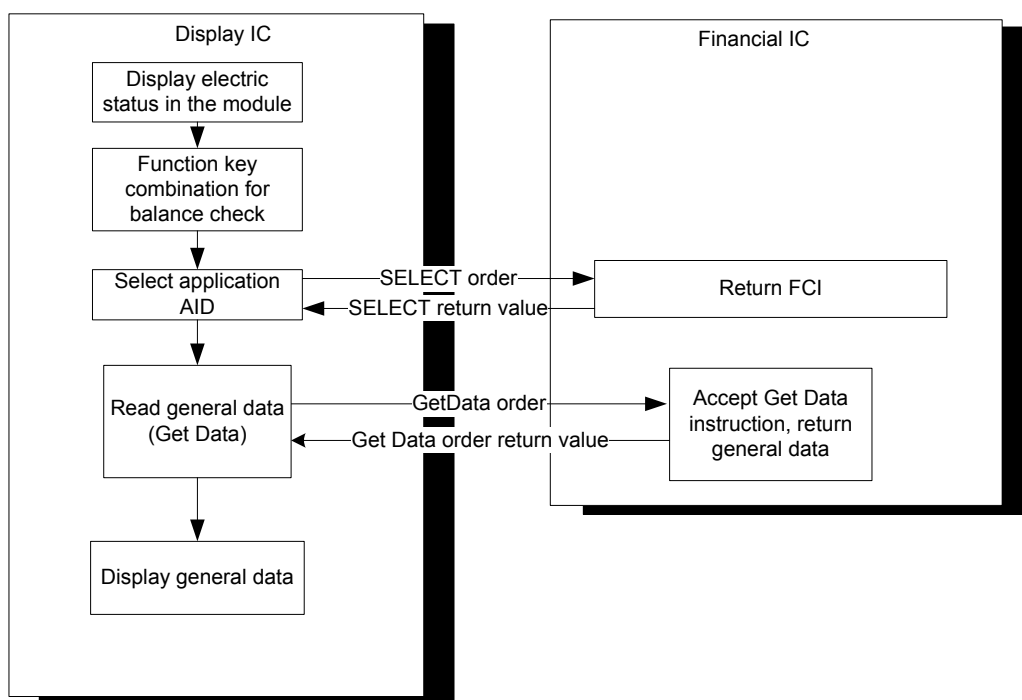


Fig. 8-3

1. If the display module is under the power-off state, after the user has pressed the power key, the display module will enter the power-on state;
2. If the display module has been under the power-on state, the user may press the key combination for general data display;
3. The display chip selects an application;
4. The display chips reads corresponding general data in the financial application by sending the Get Data command;
5. The display chip will send the general data to the display medium for display.

9 Data Elements and Commands

The exchanged commands and transmitted data elements between the financial IC chip and the display chip of financial IC display cards are defined in this part. As the transport protocol between the financial IC chip and the display chip is not defined in this Specification, this part only provides abstract definition for commands and only defines key data elements included in such commands. The data elements of the display cards must adopt TLV format. If a data element has been defined in UICS, the tag, length and format of the data element must be in conformity with requirements in UICS.

9.1 Format of Data Elements

The data elements in this part are used for calling corresponding display applications within the card, and provide display output and keyboard input functions for internal applications of the display card.

9.1.1 Format of Data Elements of Commands under Active Push Mode of Financial IC Chip

The data element packages actively pushed by the financial IC chip shall adopt following formats:

Name of Data Elements:	Value
Data content template tag of the display card	To be defined by the provider
Data content length of the display card (follow-up data length)	Lengthened
Data display	Adopt TLV format

Table 9-1

The display data may include multiple data fields. Each field shall adopt TVL format.

9.1.1.1 Balance Display Command and Data Elements

By sending the balance display command, the financial IC chip will actively push the EC balance in the card to the display chip.

The abstract definition of the balance display command is as follows:

Command format: balaDisplay(bala, currency, disp)

Data	Tag	Length
EC balance	9F79	6
Application currency code	9F42	2
Display mode	To be defined by the provider	1

Table 9-2

Display mode: Whether the identifier will be immediately displayed on the display medium: “01” indicates the updated data will be sent to the display module and be displayed immediately; “02” indicates the updated data will be sent to the display module while not be displayed immediately; “03” indicates the updated data will be displayed while not be stored.

9.1.2 Transaction Log Display Command and Format of Data Elements

The transaction log display command is used to send the past transaction records to the display module.

The abstract definition of the past transaction record display command is as follows:

Command format: TranLogDisplay(date,time,amount,trans_type, disp)

Suggested minimum data set:

Name of Data Elements:	Tag	Length
Transaction date	9A	3
Transaction time	9F12	3
Authorized amount	9F02	6
Transaction type	9C	1

Table 9-3

9.1.3 General Data Display Command and Format of Data Elements

The general data display command is used to send various data undefined in this Specification or self-defining data of the issuer to the display module of the card.

Command format: dataDisplay(data, disp)

Name of Data Elements:	Tag	Length
Data: data undefined in this Specification or self-defining data of the issuer	Data undefined in this Specification shall adopt 9FXX, as suggested in UICS, and shall not conflict with current Tags.	Lengthened

Table 9-4

9.2 transaction Command under Active Mode of Display Chip and Format of Data Elements

The transaction command under active mode of the display chip shall be executed according to UICS. The display chip sends commands to and accepts data from the financial chip via the contact interface in the same way as the terminal that sends the transaction data to the financial chip via the contact interface.

10 Application of CUP Dynamic Passwords

The dynamic password function of the display cards shall be in conformity with the *UnionPay Integrated Circuit Card Specification – Product Specification Part 3 Dynamic Password Specification*. The difference between the dynamic password function of the display cards and the dynamic password application specification is that the display module replaces the terminal to send commands to the financial IC:

- As the display cards have the display and keyboard input functions, the password calculation is no longer driven by the GENERATE AC command at the reader terminal; instead, the function keys on the card will send the APDU command to the financial IC via the display module to initiate the dynamic password function.
- Upon receiving the application cryptogram, the financial IC will complete the information combination and compression within the display module and display the dynamic password on the display medium. The user may upload the dynamic password to the authentication server of the issuer for dynamic password authentication.

The display time length of dynamic passwords is a continuous, configurable time range and may be configured according to the issuer's needs.

Time interval for time range: 0-128seconds

Default time length: 30seconds

The CUP dynamic password application is defined to be an optional application of the display cards.

10.1 Overview of CUP Dynamic Password Function

The CUP dynamic password application is based on the method that the financial IC card user authentication and transaction authorization is completed via cryptogram under online payment environment and it also generates the application cryptogram via the standard command-- GENERATE AC as regulated in the UICS specification. The differences of the two lie in: the financial IC card online transaction authorization (payment transaction) will require the cryptogram information data (CID), application transaction counter (ATC), application cryptogram (AC) and the issuer's application data (IAD) returned by the full uploading command; however, the financial IC card dynamic password method only requires the ATC, AC and the Card verification result (CVR) in the IAD returned by some parts of the uploading command; the concrete operation procedures are: the challenge code or transaction information input at the reader terminal will be filled into corresponding space of the source of transaction data to cover the original default data in order to differentiate the transaction processes. The changed transaction data source will be sent to the IC card as input data. Relevant cryptograms generated by the card will be fed back to the reader terminal, which will then combine and compress key in-

formation into digits displayed on the screen. The card holder may provide the digits to the issuer for verification.

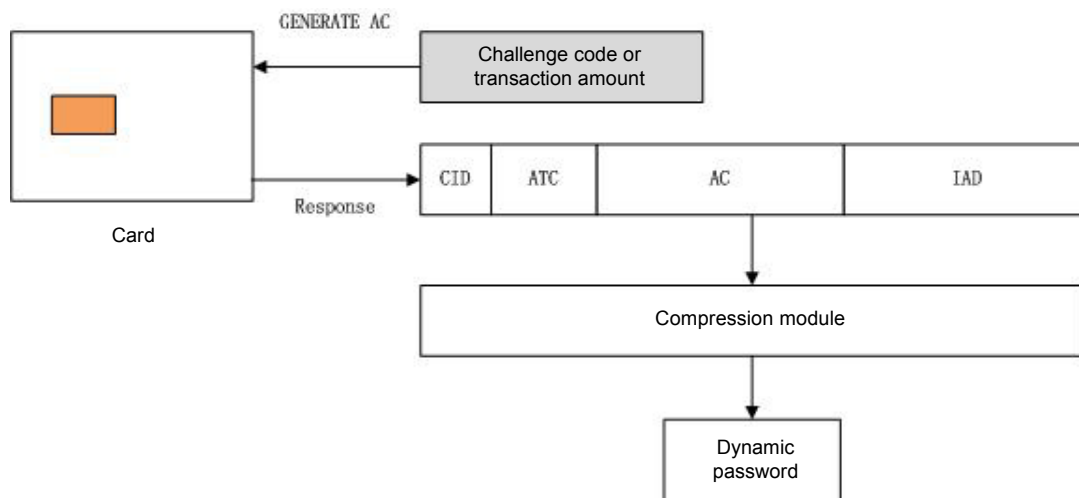


Fig. 10-1

After obtaining the dynamic password, the issuer will conduct anti-compression and repeat the encryption process for relevant information by adopting the same calculation method. If the same cryptogram is obtained, it may be proved the transaction is conducted via true cards.

10.2 Application Procedures

The basic application flow chart based on the dynamic passwords of the display cards is as follows:

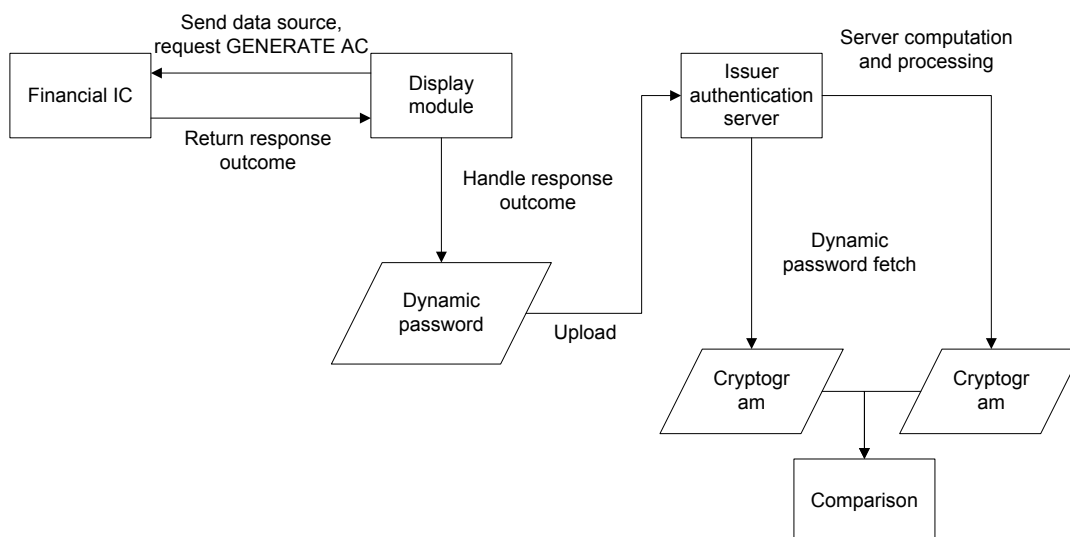


Fig. 10-2

10.3 Data Elements and Commands

The data elements, commands and the compression algorithm of the display module used in the dynamic password applications of the CUP display cards shall be in conformity with the UICS dynamic password application specification.

11 Application of Non- CUP Standard Dynamic Passwords

Non-CUP standard dynamic passwords refer to dynamic passwords generated by the display chip which have no relationship with the financial IC chip.

The application of non- CUP standard dynamic passwords is an optional application of the display cards. The application procedures, data elements and commands of non-CUP standard dynamic passwords are not defined in this Specification. The requirements on application and security are as follows:

1. The application of non-CUP standard dynamic passwords shall not influence other applications defined in this Specification;
2. The algorithm for generating dynamic passwords shall meet the requirements of commercial cryptographic algorithm.

12 Security Mechanism

12.1 Security Requirements for Display Module

Except the interface between the display module and the financial IC chip, the cards shall not have any external communication interface.

If the display chip is destroyed by external force, it must be able to destroy all internal data.

No operations endangering the security of the IC chip are allowable.

The failure of the display module shall not influence the normal operation of the financial IC chip.

12.2 Communication Security Requirements

The communication mode with check functions, such as odd-even check or CRC, shall be applicable to communications between the financial IC chip and the display module to avoid communication mistakes.

12.3 Data Security Requirements

The tamper-resistant mechanism must be available to data stored in the display chip to prevent changes to the data within the display chip, except for changes made via the financial IC interface or via built-in applications of the display chip permitted by the issuer.

It is not suggested to transmit sensitive data between the security module and the display module. If so required by the issuer, the sensitive data must be transmitted in form of signed cryptogram and shall not be stored in the display module. The transmission of non-sensitive data is not defined in this Specification.

Appendix A (Informative Annex) Layout of the Display Cards

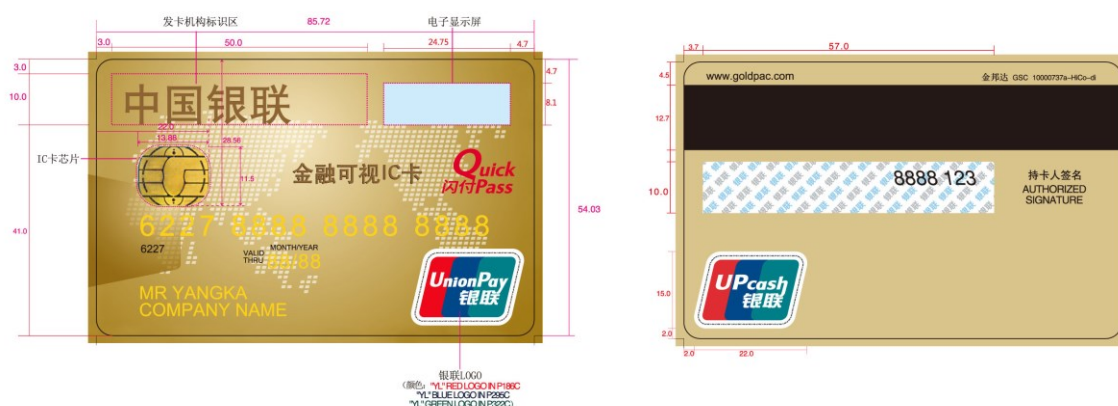
This part describes the layout of the display cards. It is a suggested choice for the issuer, instead of a compulsory requirement of this Specification.

A.1 Elements and layout parameters of the display cards without keys

The front side of the display cards without keys shall include following elements: the CUP logo, issuer's logo, card number, chip and CUP quick pass logo.

For credit cards, the effective (expiry) date, name of the card holder, fluorescent anti-counterfeit label and pre-printed BIN shall be also included.

The layout elements of the display debit cards without keys are as follows:



The layout elements of the display credit cards without keys are as follows:

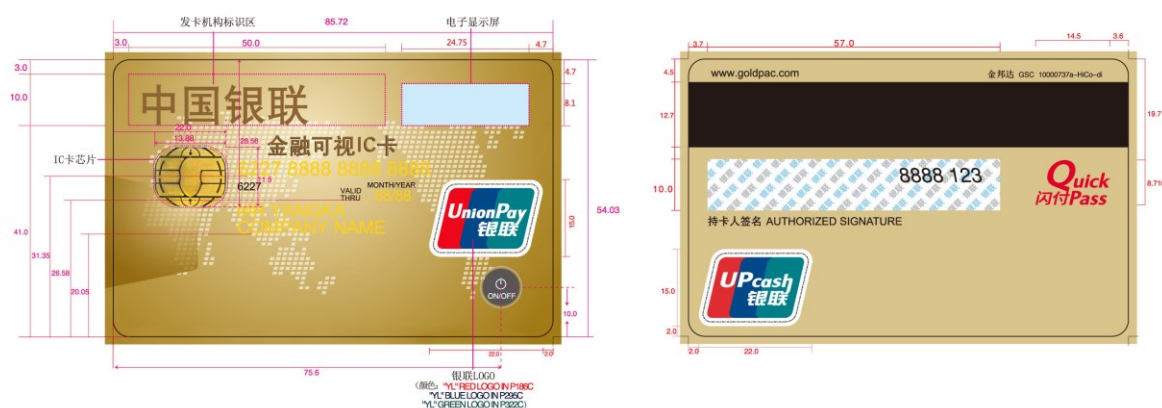


A.2 Elements and layout of the display cards with 1 key

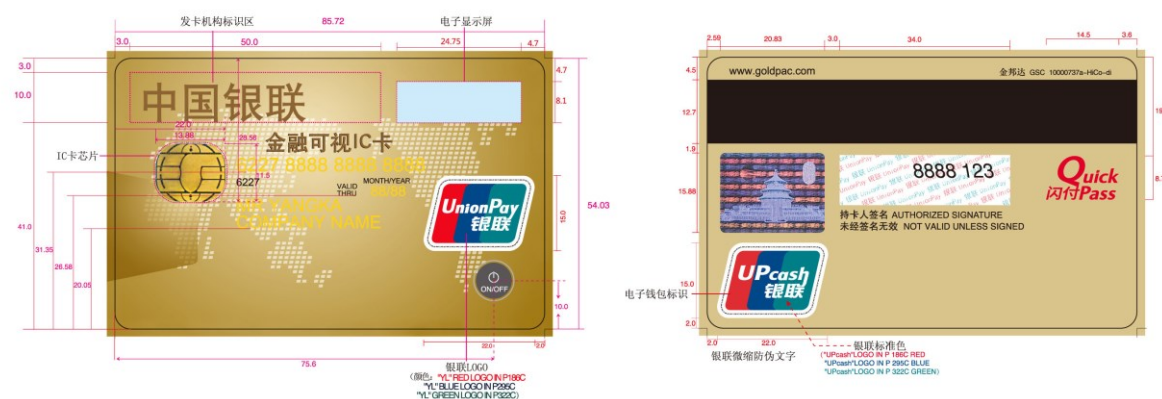
The front side of the display cards with 1 key shall include following elements: the CUP logo, issuer's logo, card number, chip, display medium and a key.

For credit cards, the effective (expiry) date, name of the card holder, fluorescent anti-counterfeit label and pre-printed BIN shall be also included.

The layout elements of the display debit cards with 1 key are as follows:



The layout elements of the display credit cards with 1 key are as follows:

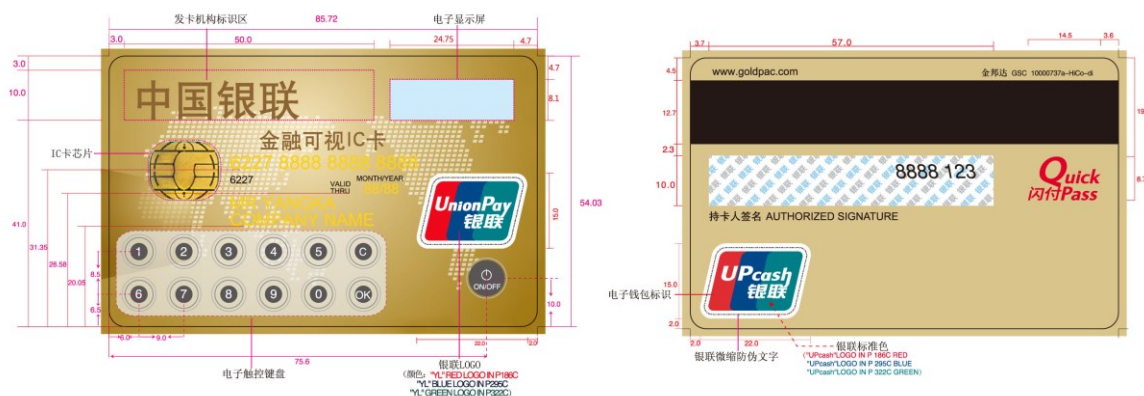


A.3 Elements and layout of the display cards with 12 key in front side

The front side of the display cards with 12 key shall include following elements: the CUP logo, issuer's logo, card number, 12 touch keys and the power key.

For credit cards, the effective (expiry) date, name of the card holder, fluorescent anti-counterfeit label and pre-printed BIN shall be also included.

The layout elements of the display debit cards with 12 key are as follows:



Layout parameters for the front side of the debit cards

Parameters	Specifications and Requirements	Tolerance
Issuer's logo area		
Height of the area	10.0mm	±0.10mm
Width of the area	50.0mm	±0.10mm
Distance between the upper frame of the area and the card's upper edge	3.0mm	±0.30mm
Distance between the left frame of the area and the card's left edge	3.0mm	±0.30mm
CUP logo		
Height of the CUP logo	15.00mm	±0.10mm
Projection width of the CUP logo	22.00mm	±0.10mm
Distance between the three-color CUP logo image and its upper and lower frames	1.60mm	±0.15mm
Background color between the three-color CUP logo image and the logo frame	White (☆2)	/
Micro-letters are printed in the middle of the three- color CUP logo image and the logo frame.	Gulim, 0.7Pt	/
Horizontal distance between the upper right external frame of the CUP logo and the card's right edge.	2.00mm	±0.30mm
Vertical distance between the lower horizontal external frame of the CUP logo and the card's lower edge.	2.00mm	±0.30mm

Parameters	Specifications and Requirements	Tolerance
Vertical distance between the lower horizontal external frame of the CUP logo and card's lower edge (applicable to cards with keys)	15.56 mm	±0.30mm
Radius of the top right corner and the left bottom corner of the CUP logo.	2.00mm	±0.10mm
Display medium		
Height of the display medium	8.1mm	±0.10mm
Width of the display medium	24.7mm	±0.10mm
Distance between the upper frame of the display medium and the card's upper edge	4.7mm	±0.30mm
Distance between the right frame of the display medium and the card's right edge	4.7mm	±0.30mm
Quick pass logo (front side)		
Height of the quick pass logo	8.716mm	±0.10mm
Width of the quick pass logo	14.5mm	±0.10mm
Distance between the upper frame of the quick pass logo and the card's upper edge	17.5mm	±0.30mm
Distance between the left frame of the quick pass logo and the card's right edge	3.6mm	±0.30mm
Contact chip		
Height of the IC card chip	11.5mm	±0.10mm

Parameters	Specifications and Requirements	Tolerance
Width of the IC card chip	13.88mm	±0.10mm
Distance between the right frame of the IC card chip and the card's left edge	22.0mm	±0.30mm
Distance between the lower frame of the IC card chip and the card's upper edge	28.56mm	±0.30mm
Electronic touch keyboard (12 keys)		
Height of the electronic touch keyboard	17.0mm	±0.10mm
Width of the electronic touch keyboard	54.05mm	±0.10mm
Distance between the left frame of the electronic touch keyboard and the card's left edge	1.46mm	±0.30mm
Distance between the lower frame of the electronic touch keyboard and the card's lower edge	2.11mm	±0.30mm
Horizontal space between key centers of the electronic touch keyboard	9.0mm	±0.10mm
Vertical space between key centers of the electronic touch keyboard	8.5mm	±0.10mm
Safety distance between the right and left frames of the electronic touch keyboard and the key area.	1.1mm	±0.10mm
Safety distance between the upper and lower frames of the electronic touch keyboard and the key area.	0.92mm	±0.10mm
Personalization area		
Bank card No.	Standard-based personalization	±0.10mm

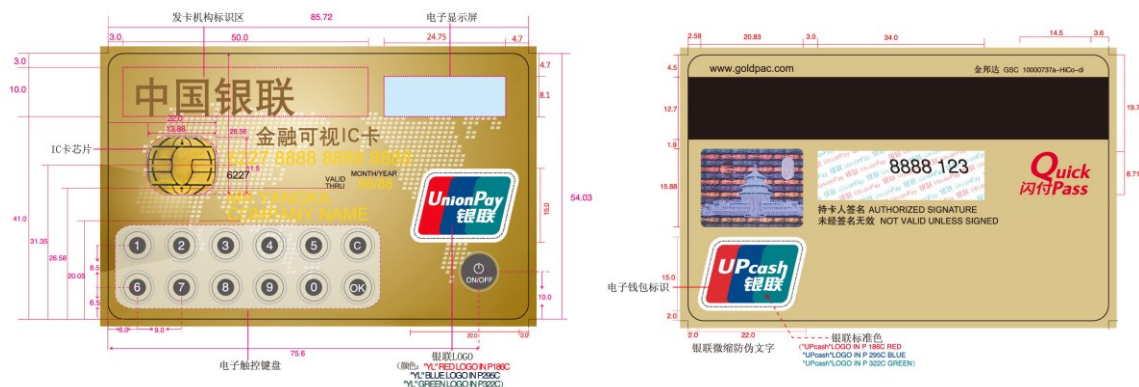
Parameters	Specifications and Requirements	Tolerance
Mechanical key (1 key)		
Height of the mechanical key	8.0mm	±0.10mm
Width of the mechanical key	8.0mm	±0.10mm
Distance between the mechanical key center and the card's left edge	75.6mm	±0.30mm
Distance between the mechanical key center and the card's lower edge	10.0mm	±0.30mm

Layout parameters for the back side of the debit cards

Parameters	Specifications and Requirements	Tolerance
Magstrip		
Height of the magstrip	12.7mm	±0.10mm
Width of the magstrip	85.72mm	±0.30mm
Distance between the upper frame of the strip and the card's upper edge	4.50mm	±0.30mm
EC logo		
Height of the EC logo	15.0mm	±0.10mm
Projection width of the EC logo	22.0mm	±0.10mm
Distance between the three-color EC logo image and its upper and lower frames	1.60mm	±0.15mm
Background color between the three-color EC logo image and the logo frame	White (☆2)	/

Parameters	Specifications and Requirements	Tolerance
EC letters are printed in the middle of the new three- color logo image and the logo frame.	Gulim, 0.7Pt	/
Horizontal distance between the left bottom external frame of the EC logo and the card's left edge.	2.00mm	±0.30mm
Vertical distance between the lower horizontal external frame of the EC logo and card's lower edge	2.00mm	±0.30mm
Radius of the top right corner and the left bottom corner of the EC logo	2.00mm	±0.30mm
Signature strip		
Width of the signature strip	57.0mm	±0.10mm
Height of the signature strip	10.0mm	±0.10mm
Distance between the left frame of the signature strip and the card's left edge	3.7mm	±0.30mm
Distance between the upper frame of the signature strip and the card's upper edge	19.5mm	±0.30mm
Quick pass logo (back side)		
Height of the quick pass logo	8.716mm	±0.10mm
Width of the quick pass logo	14.5mm	±0.10mm
Distance between the upper frame of the quick pass logo and the card's upper edge	19.7mm	±0.30mm
Distance between the left frame of the quick pass logo and the card's left edge	3.6mm	±0.30mm

Layout parameters for credit cards:



Layout parameters for the front side of the credit cards:

Parameters	Specifications and Requirements	Tolerance
Issuer's logo area		
Height of the area	10.0mm	±0.10mm
Width of the area	50.0mm	±0.10mm
Distance between the upper frame of the area and the card's upper edge	3.0mm	±0.30mm
Distance between the left frame of the area and the card's left edge	3.0mm	±0.30mm
CUP logo		
Height of the CUP Logo	15.00mm	±0.10mm
Projection width of the CUP logo	22.00mm	±0.10mm
Distance between the three-color CUP logo image and its upper and lower frames	1.60mm	±0.15mm
Background color between the three-color CUP logo image and the logo frame	White (☆2)	/

Parameters	Specifications and Requirements	Tolerance
Micro-letters are printed in the middle of the new three- color CUP logo image and the logo frame.	Gulim, 0.7Pt	/
Horizontal distance between the upper right external frame of the CUP logo and the card's right edge.	2.00mm	±0.30mm
The vertical distance between the lower horizontal external frame of the CUP logo and card's lower edge	2.00mm	±0.30mm
Radius of the top right corner and the left bottom corner of the CUP logo.	2.00mm	±0.10mm
Display medium		
Height of the display medium	8.1mm	±0.10mm
Width of the display medium	24.75mm	±0.10mm
Distance between the upper frame of the display medium and the card 's upper edge	4.7mm	±0.30mm
Distance between the right frame of the display medium and the card's right edge	4.7mm	±0.30mm
Quick pass logo (front side)		
Height of the quick pass logo (back side)	8.716mm	±0.10mm
Width of the quick pass logo	14.5mm	±0.10mm
Distance between the upper frame of the quick pass logo and the card's upper edge	17.5mm	±0.30mm

Parameters	Specifications and Requirements	Tolerance
Distance between the left frame of the quick pass logo and the card's right edge	3.6mm	±0.30mm
Contact chip		
Height of the IC card chip	11.5mm	±0.10mm
Width of the IC card chip	13.88mm	±0.10mm
Distance between the right frame of the IC card chip and the card's left edge	22.0mm	±0.30mm
Distance between the lower frame of the IC card chip and the card's upper edge	28.56mm	±0.30mm
Electronic touch keyboard (12 keys)		
Height of the electronic touch keyboard	17.0mm	±0.10mm
Width of the electronic touch keyboard	54.05mm	±0.10mm
Distance between the left frame of the electronic touch keyboard and the card's left edge	1.46mm	±0.30mm
Distance between the lower frame of the electronic touch keyboard and the card's lower edge	2.11mm	±0.30mm
Horizontal space between the central points of keys of the electronic touch keyboard	9.0mm	±0.10mm
Vertical space between the central points of keys of the electronic touch keyboard	8.5mm	±0.10mm
Safety distance between the right and left frames of the electronic touch keyboard and the key area.	1.1mm	±0.10mm

Parameters	Specifications and Requirements	Tolerance
Safety distance between the upper and lower frames of the electronic touch keyboard and the key area.	0.92mm	±0.10mm
Personalization area		
Bank's card No.(except for 12 keys)	Standard-based personalization	±0.10mm
Pre-printed BIN No. (except for 12 keys)	Standard-based personalization	±0.10mm
Effective date (except for 12 keys)	Standard-based personalization	±0.10mm
Authorized signature (except for 12 keys))	Standard-based personalization	±0.10mm
Mechanical key (1 key)		
Height of the mechanical key	8.0mm	±0.10mm
Width of the mechanical key	8.0mm	±0.10mm
Distance between the mechanical key center and the card's left edge	75.6mm	±0.30mm
Distance between the mechanical key center and the card's lower edge	10.0mm	±0.30mm

Note 2: the bank card number shall be photo-etched.

Layout parameters for the back side of the credit cards:

Parameters	Specifications and Requirements	Tolerance
Magstrip		

Parameters	Specifications and Requirements	Tolerance
Height of the magstrip	12.7mm	±0.10mm
Width of the magstrip	85.72mm	±0.30mm
Distance between the top frame of the area and the card's upper edge	4.50mm	±0.30mm
EC logo		
Height of the EC logo	15.0mm	±0.10mm
Projection width of the EC logo	22.0mm	±0.10mm
Distance between the three-color EC logo image and its upper and lower frames	1.60mm	±0.15mm
Background color between the three-color EC logo image and the logo frame	White (☆2)	/
EC letters are printed in the middle of the new three- color logo image and the logo frame.	Gulim, 0.7Pt	/
Horizontal distance between the left bottom external frame of the CE logo and the card's left side.	2.00mm	±0.30mm
The vertical distance between the lower horizontal external frame of the CE logo and card's lower edge	2.00mm	±0.30mm
Radius of the top right corner and the left bottom corner of the EC logo	2.00mm	±0.30mm
CUP hologram		
Width of the CUP hologram	20.83mm	±0.10mm

Parameters	Specifications and Requirements	Tolerance
Height of the CUP hologram	15.88mm	$\pm 0.10\text{mm}$
Distance between the left frame of the CUP hologram and the card's left edge	2.59mm	$\pm 0.30\text{mm}$
Distance between the upper frame of the CUP hologram and the card's upper edge	19.1mm	$\pm 0.30\text{mm}$
Signature strip of the credit card		
Width of the signature strip of the credit card	34.0mm	$\pm 0.10\text{mm}$
Height of the signature strip of the credit card	10.0mm	$\pm 0.10\text{mm}$
Distance between the left frame of the signature strip and the card's left edge	26.4mm	$\pm 0.30\text{mm}$
Distance between the upper frame of the signature strip and the card's upper edge	19.5mm	$\pm 0.30\text{mm}$
Quick pass logo (back side)		
Height of the quick pass logo	8.716mm	$\pm 0.10\text{mm}$
Width of the quick pass logo	14.5mm	$\pm 0.10\text{mm}$
Distance between the upper frame of the quick pass logo and the card edge	19.7mm	$\pm 0.30\text{mm}$
Distance between the left frame of the quick pass logo and the card's left edge	3.6mm	$\pm 0.30\text{mm}$